Topic 2 – States of matter and mixtures

States of matter

Students should:		Maths skills
2.1	Describe the arrangement, movement and the relative energy of particles in each of the three states of matter: solid, liquid and gas	5b
2.2	Recall the names used for the interconversions between the three states of matter, recognising that these are physical changes: contrasted with chemical reactions that result in chemical changes	
2.3	Explain the changes in arrangement, movement and energy of particles during these interconversions	5b
2.4	Predict the physical state of a substance under specified conditions, given suitable data	1d 4a

Use of mathematics

- Translate information between diagrammatic and numerical forms (4a).

Methods of separating and purifying substances

Students should:		Maths skills
2.5	Explain the difference between the use of 'pure' in chemistry compared with its everyday use and the differences in chemistry between a pure substance and a mixture	
2.6	Interpret melting point data to distinguish between pure substances which have a sharp melting point and mixtures which melt over a range of temperatures	1a
2.7	Explain the types of mixtures that can be separated by using the following experimental techniques: a simple distillation b fractional distillation c filtration d crystallisation e paper chromatography	
2.8	Describe an appropriate experimental technique to separate a mixture, knowing the properties of the components of the mixture	

Students should:		Maths skills
2.9	Describe paper chromatography as the separation of mixtures of soluble substances by running a solvent (mobile phase) through the mixture on the paper (the paper contains the stationary phase), which causes the substances to move at different rates over the paper	
2.10	Interpret a paper chromatogram: a to distinguish between pure and impure substances b to identify substances by comparison with known substances c to identify substances by calculation and use of R_f values	3a, 3c 4a
2.11	*Core Practical: Investigate the composition of inks using simple distillation and paper chromatography*	
2.12	Describe how: a waste and ground water can be made potable, including the need for sedimentation, filtration and chlorination b sea water can be made potable by using distillation c water used in analysis must not contain any dissolved salts	

Use of mathematics

- Interpret charts (4a).